

European Commission

MLE on National Policies for AI in Science

Horizon Europe

PSF CHALLENGE - MUTUAL LEARNING EXERCISE

PSF supports national policies for AI in science

Artificial Intelligence (AI) is rapidly transforming science by accelerating discovery, automating research tasks, and enabling new paradigms of investigation. The growing role of AI in science has placed it high on the European political agenda, including the commitment to establish an AI in Science Strategy and a European AI Research Council in the form of a Resource for AI Science in Europe (RAISE), as reflected in the Political Guidelines for 2024-2029. This commitment was further underlined in the AI Continent Action Plan and in the mandate of the Commissioner for Research, Innovation and Startups, highlighting the EU's ambition to lead in this area.

However, the development and adoption of AI for science strategies varies across EU Member States and Associated Countries of Horizon Europe. Countries are at different stages of readiness, facing common challenges but responding with different policy approaches. Although national contexts vary widely, there is a shared recognition of AI's transformative potential for science, and of the urgent need for strategic, coordinated policy responses.

To explore how national policies can effectively support the implementation of AI in scientific research, the Horizon Europe Policy Support Facility (PSF) launched a Mutual Learning Exercise (MLE) on National Policies for AI in Science. The MLE aimed to foster mutual learning, assess national capabilities and develop actionable recommendations for policy innovation at national and institutional levels.

About the Mutual Learning Exercise

The MLE on National Policies for AI in Science took place from July 2024 to June 2025. Eleven countries participated: Austria, Belgium, Estonia, Germany, Hungary, Ireland, Italy, the Netherlands, Norway, Romania and Slovenia. Three in-depth country visits were held in Brussels, Oslo and Pisa, each addressing one of the MLE's core themes: (1) framework conditions and funding for AI in science; (2) infrastructures for AI in science; and (3) talent for AI in science.

Diverse national landscapes, common barriers

Despite national differences, several shared challenges emerged from the exercise. These include limited access to high-performance computing and data infrastructures; fragmented and underused scientific data and poor interoperability; critical skills shortages and difficulties in retaining interdisciplinary talent. Governance frameworks also often lag behind scientific and technological developments and ethical principles remain difficult to operationalise.

Additionally, the lack of coordination across government departments and the absence of clear institutional incentives to promote AI experimentation are slowing progress. These barriers risk reinforcing disparities in scientific capacity and innovation potential across Europe.

A menu of policy recommendations

A set of policy recommendations for national policymakers and research organisations were developed along the three core themes of the MLE. The recommendations are intended to be flexible and adaptable to each country's context and stage of AI maturity in science- whether emerging, expanding or advanced.

On the topic of **funding and framework conditions**, national policymakers need to embed AI in science into their national strategies and roadmaps through dedicated funding programmes and incentives for interdisciplinary collaboration. This includes aligning AI strategies with ERA objectives, setting up anticipatory governance structures, and providing coherent policies across departments. Research organisations are encouraged to create protected spaces and fellowships to support early experimentation with AI, align institutional strategies with national and EU priorities, and foster communities of practice that promote interdisciplinary collaboration.

In relation to **infrastructures**, national authorities are advised to support accessible, interoperable and sustainable infrastructures, including domain-specific AI systems, high-performance computing and secure cloud-based data environments. Strategic engagement with EU-level initiatives such as EuroHPC and the European Open Science Cloud are also recommended. At institutional level, research organisations should ensure that infrastructure is efficiently used, FAIR (Findable, Accessible, Interoperable, and Reusable) data practices are promoted, researchers are trained in data management and AI validation, and technical support roles such as data stewards and AI advisors are provided.

In terms of **talent**, national governments are encouraged to introduce AI education in school curricula, fund interdisciplinary AI training and research programmes, and create structured career paths and co-financing schemes with industry. Research organisations should expand AI education across scientific disciplines, reform the evaluation criteria to reward interdisciplinary and AI-related scientific activities, and offer training in responsible AI use. Support for interdisciplinary PhD and AI-specialised postdoctoral programmes was also highly recommended.

What makes AI in science policy effective?

Some challenges appeared in all three topics. Therefore, the MLE identified several overarching success factors for effective AI-in-science policy beyond the thematic recommendations. These include combining strategic,

long-term investment with flexible mechanisms that support bottom-up experimentation, and public-private partnerships. Building both technical infrastructure and human capacity – including interdisciplinary researchers, software engineers, knowledge brokers – is key to effective development.

Bridging the gap between abstract ethical principles and practical implementation also emerged as a priority during the discussions. Policies must provide researchers with clear tools, training and support to address ethical AI use in daily scientific work. Anticipatory governance mechanisms – such as foresight studies, stakeholder engagement initiatives and iterative policy cycles – could also help ensure that policy remains responsive to rapidly evolving AI technologies and practices.

Conclusion

The MLE on National Policies for AI in Science revealed a shared recognition of the transformative potential of AI for scientific research and a strong desire among Member States and Associated Countries to cooperate. Despite differing national contexts, there is clear momentum for coordinated policy responses to harness AI responsibly and effectively in science.

Readers are encouraged to consult the thematic and final reports on the PSF website for further recommendations, good practice examples and challenges encountered.

For further information:

Thematic Report 1: Framework conditions and funding for AI in Science

Thematic Report 2: Infrastructures for AI in Science

Thematic Report 3: Talent for AI in Science

Final Report

Visit the PSF website to learn more

The PSF's objective is to help Member States and Associate Countries to 'improve the design, implementation and evaluation of R&I policies.' The PSF provides expertise and practical support to Member States in three major ways: PSF Country (formerly PSF Peer Reviews & Specific support to countries); PSF Challenge (including the PSF Mutual Learning Exercises, focused on specific and operational R&I challenges of interest in several volunteering countries); and PSF Open (which allows countries that have already benefited from a PSF exercise to receive support to follow up on PSF recommendations). The PSF is funded under Horizon Europe.

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